Q1 (2 Marks Each)

1. Differentiate supervised / unsupervised learning.
2. What is Reinforcement Learning? Explain the significance of award and action in RL.
3. Explain linear separable and non linearly separable pattern with example.
4. Discuss the applications of ML.
5. What is neural network architecture? Explain logistic sigmoid function with example.
6. What are the important properties of activation function used in neural network?
7. Explain basic McCulloch Pitts neuron model with examples.
8. Differentiate between machine learning and deep learning. Applications of DL.
9. What is meant by ensemble learning? What are the different types of ensemble classifiers?
10. Explain Accuracy, Precision, and Recall, specificity and F1 score.
11. Explain the confusion matrix with examples.
12. What are the autoencoders? Discuss types of autoencoders. List applications of autoencoders.
13. Explain RNN and its uses in NLP.
14. Write short note trends in data science for Audio.
15. Explain data Science for Multimodal applications.

Q2 (5 Marks Each)

1. Implement binary gates [OR, AND, NOR, XOR] using McCulloch Pitts ANN.
2. Explain all the activation functions with neat diagram.
3. Explain Bagging and Boosting in details.
4. Explain LSTM in detail.
5. Draw and explain architecture of convolutional neural networks. Explain kernel and pooling.
6. Explain how different data modalities (text, audio, image, video) are integrated to improve the performance of user experience of multimodel system, explain with real time application.

Q3 (5 Marks Each)

1. If TP=....., FP=....., FN=....., TN=..... then what is Accuracy, precision, error rate, sensitivity (recall), Specificity and F1 Score of the model?
2. Evaluate performance of classifier1 and classifier2 on the basis of the above evaluation parameters, given the following confusion matrix, where:
   F = actual fraud, F' = predicted fraud, N = actual no. fraud and N' = predicted no. fraud.

Classifier 1

	F'	**N'**
F	20	10
N	10	60

Classifier 2

	F'	N'
F	0	15
N	5	80